

The SIGNAL Group Collaboration Network

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The SIGNAL Group Collaboration Network is an interactive visualization of co-authorship within Prof. Soudeep Deb's research group at IIM Bangalore (Decision Sciences Area). It is built from papers listed on the group's research page, covering work from 2021 through the latest preprints. The current version (October 2026) includes 35 papers and 47 people, grouped as the principal investigator, current members, alumni, and external collaborators.

Unlike a conventional co-authorship network, the default view represents each paper as a hyperedge: a single relation connecting all of its authors. This preserves the multi-author structure of a paper rather than decomposing it into separate pairwise links. For example, a four-author paper remains one collaborative event instead of being represented as six pairwise edges. Users can also switch to a pairwise projection, where edge weights record the number of shared papers, or to a bipartite author-paper graph, allowing direct comparison of what the different representations emphasize.

The visualization supports filtering by year, research theme, and individual author. Selecting a person narrows the graph to that person's papers and collaborators, while the evolution mode reveals how the group's collaboration structure has developed over time. Together, these views provide an interactive way to explore the group's publication history, collaboration patterns, and research connections.

A short note on relevant statistical concepts will be made available soon.